

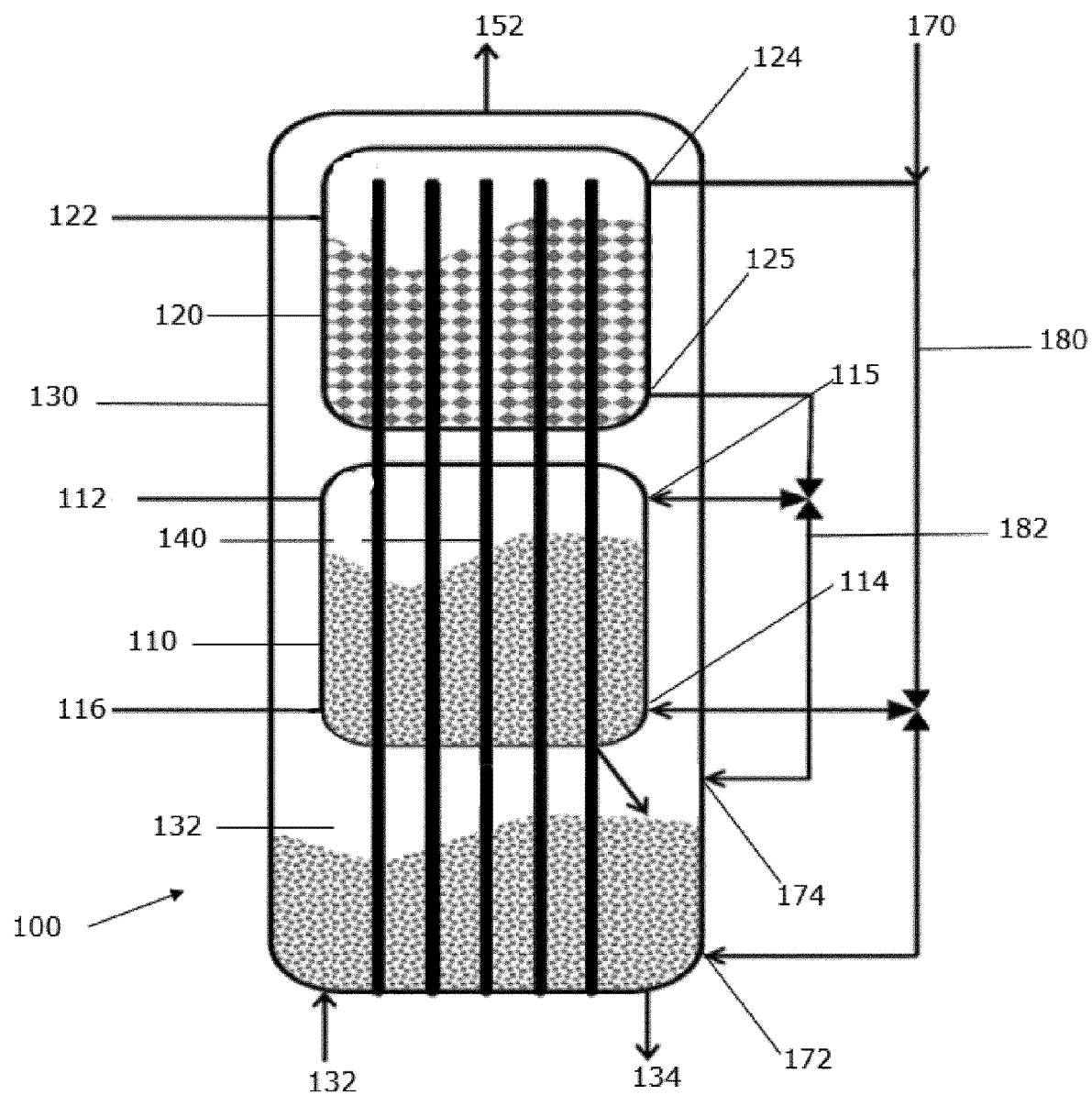


FIG. 1

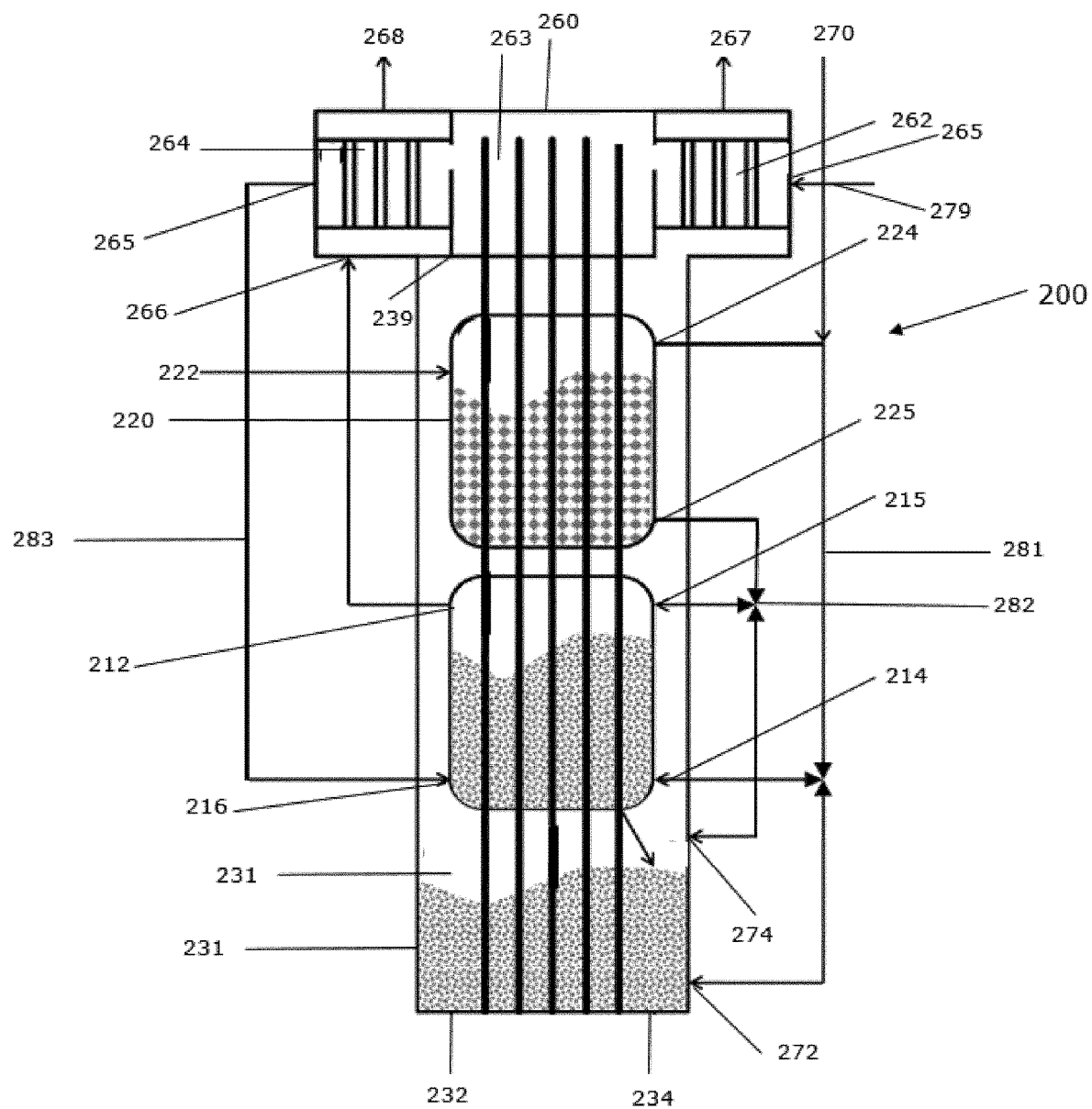


FIG. 2

A DEVICE FOR GASIFICATION OF FEEDSTOCK

TECHNICAL FIELD

[0001] The present disclosure relates to a device for gasification of feedstock.

BACKGROUND OF THE INVENTION

[0002] Gasification devices are used for gasifying feedstock. The feedstock may be biodegradable material that comprises a composition of carbon or carbohydrates containing substance, Hydrogen, Nitrogen, Calcium etc. The biodegradable material or biomass composition is generally found in organic materials like wood, plastic, pesticides, herbicides, pathogens, paints, contaminated solvents, residues from the paper and cellulose production, coal, tar, tar sand to name a few. This kind of feedstock is collected from homes, hospitals, power plants, oil refineries etc. The gasification devices make use of the feedstock to produce fuel in an environmentally friendly manner. The gasification devices produce fuel like a synthesis gas or a producer gas. Synthesis gas (Syngas) comprises carbon monoxide (CO) and hydrogen (H₂), which is the product of steam or oxygen gasification. Producer gas is the mixture of gases produced by the gasification of organic material such as biomass. Producer gas is composed of carbon monoxide (CO), hydrogen (H₂), oxygen (O₂), carbon dioxide (CO₂) and typically a range of hydrocarbons such as methane (CH₄) with nitrogen from the air. This fuel has applications including, but not limited to, operating gas, steam, hydrogen engines. **[0003]** EP1187892 discloses a facility for producing combustible gas from carbon-containing, in particular biogenic feed materials by allothermic steam gasification. The facility having the following features a pressure-supercharged fluidized-bed gasification chamber with a pressure-tight lock for supplying the feed materials, which are to be gasified. The facility is further having a filter chamber, which is connected to the fluidized-bed gasification chamber via a connecting channel. The facility is further having an external heat source. The facility is further having a heat-pipe arrangement, which takes up heat from the external heat source and gives it off to the gasification bed in the fluidized-bed gasification chamber.

Overview

[0004] Thus, there is a need to overcome drawbacks of prior art. In particular, there is a need to overcome drawbacks of conventional gasification devices. The various aspects of the present invention overcome drawbacks of the prior art. The following presents a simplified overview in order to provide a basic understanding of one or more aspects of the invention. This overview is not an extensive overview of the invention, and is neither intended to identify key or critical elements of the invention, nor to delineate the scope thereof. Rather, the primary purpose of the overview is to present some embodiments of the invention.

[0005] According to the invention in an aspect a device for gasification of feedstock is provided. The device comprises an exothermic chamber, a first endothermic chamber and a second endothermic chamber. The exothermic chamber is provided with a combustion zone that is configured to produce combustion process heat. The first endothermic chamber is adapted to perform a gasification process, in

particular, an allothermic gasification process. The second endothermic chamber is adapted to have the feedstock perform endothermic break-down reactions. The device is configured such the second endothermic chamber and the first endothermic chamber use the process heat. The device further comprises at least one heat-transfer rod adapted to transfer combustion process heat from the combustion zone of the exothermic chamber to the first endothermic chamber and to the second endothermic chamber. At least one effect can be that the rods are better than hollow tubes and more particularly, the rods are easy to maintain, high durability and better heat transfer properties.

[0006] In some embodiments, the at least one heat-transfer rod extends into the first endothermic chamber and into the second endothermic chamber. In some embodiments, the at least one heat-transfer rod is arranged inside the exothermic chamber. In some embodiments, the at least one heat-transfer rod, arranged inside the exothermic chamber, extends into the first endothermic chamber and into the second endothermic chamber.

[0007] In some embodiments, the at least one heat-transfer rod comprises a solid heat pipe, a hollow heat pipe or a convection heat pipe.

[0008] In some embodiments, at least two of the second endothermic chamber, the first endothermic chamber and the exothermic chamber are arranged above one another.

[0009] In some embodiments, at least one of the second endothermic chamber and the first endothermic chamber is arranged inside the exothermic chamber.

[0010] In some embodiments, the first endothermic chamber is a reformer.

[0011] In some embodiments, the reformer comprises a fluidized reformer-bedchamber.

[0012] In some embodiments, the first endothermic chamber includes a reformer inlet configured to be coupled to a source of steam.

[0013] In some embodiments, the device further comprising a steam generator configured to receive water and to generate steam from the water. In some embodiments, the steam generator comprises a steam outlet connected to the reformer inlet. In some embodiments, the device further comprising a steam generator configured to receive water and to generate steam from the water, wherein the steam generator comprises a steam outlet connected to the reformer inlet.

[0014] In some embodiments, the at least one heat-transfer rod is further adapted to transfer the combustion process heat from the exothermic chamber to the steam generator.

[0015] In some embodiments, the second endothermic chamber is a pyrolysis reactor.

[0016] In some embodiments, the pyrolysis reactor comprises a fluidized reactor-bedchamber.

[0017] In some embodiments, the combustion zone of the exothermic chamber is configured to combust fuel to produce the process heat.

[0018] In some embodiments, the exothermic chamber is configured to combust feedstock. In some embodiments, the exothermic chamber comprises an inlet configured to receive feedstock.

[0019] In some embodiments, the exothermic chamber further comprises a bottom inlet configured to be coupled to an oxygen source. In some embodiments, the exothermic chamber further comprises a bottom outlet configured to release ash from the exothermic chamber. In some embodi-

ments, the exothermic chamber further comprises a bottom inlet configured to be coupled to an oxygen source and a bottom outlet configured to release ash from the exothermic chamber.

[0020] The independent claim defines the invention in one aspect. The dependent claims state selected elements of embodiments according to the invention. It is to be noted that elements of these embodiments may be combined with each other unless specifically noted to the contrary.

[0021] This overview is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. This overview is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. Those skilled in the art will recognise additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a drawing that schematically illustrates a sectional view of a gasification device according to some embodiments.

[0023] FIG. 2 is a drawing that schematically illustrates a sectional view of a gasification device according to some embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0024] Below, embodiments, implementations and associated effects are disclosed with reference to the drawings that illustrate views of some embodiments. It should be noted that views of exemplary embodiments are merely to illustrate selected features of the some embodiments. In particular, cross-sectional views are not drawn to scale and dimensional relationships of the illustrated structures can differ from those of the illustrations. As used herein, like terms refer to like elements throughout the description.

[0025] It is to be understood that the features of various embodiments described herein may be combined with each other, unless specifically noted otherwise. In some instances, well-known features are omitted or simplified to clarify the description of the exemplary implementations. The order in which the embodiments/implementations and methods/processes are described is not intended to be construed as a limitation, and any number of the described implementations and processes may be combined.

[0026] FIG. 1 is a drawing that schematically illustrates a sectional view of a gasification device 100 for gasification of feedstock according to an aspect of the invention. In some embodiments, the gasification device 100 comprises an exothermic chamber 130, a first endothermic chamber 110 and a second endothermic chamber 120. In some embodiments (not shown), the first endothermic chamber comprises the second endothermic chamber. In some embodiments (not shown), the device comprises an exothermic chamber and a main endothermic chamber that comprises a first endothermic chamber and a second endothermic chamber. At least one effect can be that steps of a gasification process can take place within the gasification device 100, i.e., performing endothermic break-down reactions on feedstock to produce a gas and performing exothermic reactions to produce

process heat, wherein the process heat can be used for carrying out the endothermic reactions.

[0027] In some embodiments, the gasification device 100 has at least two of the second endothermic chamber 120, the first endothermic chamber 110 and the exothermic chamber 130 arranged above one another. At least one effect can be that heat transfer by convection is improved in comparison with alternate arrangements. Further, gravity can be used in providing feedstock to the first endothermic chamber 110, to the second endothermic chamber 120 and/or to the exothermic chamber 130. Thus, the efficiency of the gasification device 100 can be improved. Further effect can be that the gasification device 100 is kept structurally simple. In particular, the gasification device 100 can efficiently allow the material to flow, reducing loss of heat or resources to the surrounding, from the second endothermic chamber 120 to the first endothermic chamber 110, from the second endothermic chamber 120 to the exothermic chamber 130, and/or from the first endothermic chamber 110 to the exothermic chamber 130.

[0028] In some embodiments, the gasification device 100 has at least one of the first endothermic chamber 110 and the second endothermic chamber 120 arranged inside the exothermic chamber 130. In some embodiments, the gasification device 100 has both the first endothermic chamber 110 and the second endothermic chamber 120 arranged inside the exothermic chamber 130. In some embodiments, the exothermic chamber 130 is configured to have the process heat produced inside the exothermic chamber 130 by combustion of feedstock. At least one effect can be that the first endothermic chamber 110 and the second endothermic chamber 120 carry out the respective endothermic reactions using process heat produced by the exothermic chamber 130. Thus, the need for an external heat source is reduced. Accordingly, the first endothermic chamber 110 can be configured to perform endothermic reactions without use of any additional exothermic chamber. Likewise, the second endothermic chamber 120 can be configured to perform endothermic reactions without use of any additional exothermic chamber.

[0029] The gasification device 100 for gasification of feedstock further comprises a plurality of heat-transfer rods 140. In one embodiment (not shown), the gasification device comprises a single heat-transfer rod. In some embodiments, the plurality of heat-transfer rods 140 extend from the combustion zone of the exothermic chamber 130 to the first endothermic chamber 110. Thus, the plurality of heat-transfer rods 140 can transport process heat generated in the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110. In some embodiments, the plurality of heat-transfer rods 140 extend from the combustion zone of the exothermic chamber 130 to the second endothermic chamber 120. Thus, the plurality of heat-transfer rods 140 can transport process heat generated in the combustion zone 131 of the exothermic chamber 130 to the second endothermic chamber 120. In some embodiments, the plurality of heat-transfer rods 140 extends from the combustion zone of the exothermic chamber 130 through the first endothermic chamber 110 to the second endothermic chamber 120. Thus, the plurality of heat-transfer rods 140 can transport process heat generated in the combustion zone 131 of the exothermic chamber 130 to the second endothermic chamber 120 and to the first endothermic chamber 110.

[0030] In the example illustrated in FIG. 1, the gasification device 100 is configured such that both, the second endothermic chamber 120 and the first endothermic chamber 110, use the process heat produced in the exothermic zone 131 of the exothermic chamber 130. The process heat, generated in the combustion zone 131 of the exothermic chamber 130 and produced at a temperature of at least 1100° C., for example, at a temperature in a range of from 1100° C. to 1600° C., in particular at a temperature in a range of from 1200° C. to 1400° C., rises up to the first endothermic chamber 110 and then to the second endothermic chamber 120. In some embodiments, the gasification device 100 is configured to additionally use convection, wherein, for example, hot gas rises from the combustion zone 131 of the exothermic chamber 130 along an outside surface of a wall of the first endothermic chamber 110 to exchange heat with the wall of the first endothermic chamber 110 and transfer energy to the first endothermic chamber 110. Likewise, the hot gas can rise further from the first endothermic chamber 110 along an outside surface of a wall of the second endothermic chamber 120. Thus, a first portion of the process heat is transported to the first endothermic chamber 110 operating at the temperature of at least 700° C. As the process heat rises up in the gasification device 100, heat transport takes place to the outside of the device and the temperature of the process heat decreases. Further, a second portion of the process heat is transported to the second endothermic chamber 120 operating at the temperature of at least 350° C. Thus, process heat produced in the combustion zone 131 of the exothermic chamber 130 heats the outside of both, the first endothermic chamber 110 and the second endothermic chamber 120. Some of the process heat reaching the second endothermic chamber 120 enters into the performance of pyrolysis.

[0031] At least one effect of the above configuration can be that the device can be configured so as to achieve heat transfer as required. The gasification device 100 achieves an improved heat transfer from the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110 and/or to the second endothermic chamber 120, since it uses at least two paths of heat transfer: A first path is provided by convective heat transfer that takes place from the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110 and/or to the second endothermic chamber 120. A second path is provided by conductive heat transfer from the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110 and/or to the second endothermic chamber 120 that takes place heat transfer in the at least one heat-transfer rod 140 that extends from the combustion zone 131 of the exothermic chamber 130 through the inside of the first endothermic chamber 110 to the inside of the second endothermic chamber 120.

[0032] Having regard to the exothermic chamber 130 of the gasification device 100, the combustion zone 131 is adapted to produce process heat in the combustion zone 131. In the combustion zone, the exothermic chamber 130 is configured to withstand a temperature that can occur during combustion. For example, the exothermic chamber 130 is configured to withstand a temperature of 800° C., of 900° C., or of 1000° C. In some embodiments, the exothermic chamber 130 is configured to withstand a temperature of 1100° C.

[0033] In some embodiments, the exothermic chamber 130 is configured to receive the feedstock for combustion in the form of pyrolysis vapors, gases and/or pyrolysis coke or ash from the first endothermic chamber 110 and/or from the second endothermic chamber 120 and/or directly from an external source 170 to produce process heat.

[0034] The exothermic chamber 130 comprises a first inlet 172 adapted to receive feedstock for combustion in the combustion zone 131 from an external source of feedstock. Further, in one example, a second inlet 174 is provided to the exothermic chamber 130 in proximity to the combustion zone 131. The second inlet 174 is adapted to receive feedstock from the first endothermic chamber 110 and/or from the second endothermic chamber 120. In some embodiments (not shown), the exothermic chamber 130 has one inlet to commonly receive feedstock for combustion from any one or any combination of an external source, the first endothermic chamber 110 and/or the second endothermic chamber 120. Thus, feedstock can continuously be supplied to the exothermic chamber 130 for combustion in the combustion zone 131. The exothermic chamber 130 can be configured to receive the feedstock for combustion as intermediate products in any form, like a mixture of pyrolysis vapors, gases and/or pyrolysis coke or ash, from the first endothermic chamber 110 and/or from the second endothermic chamber 120 to produce process heat. Thus, feedstock can continuously be supplied to the combustion zone 131 for combustion in the exothermic chamber 130 to produce process heat.

[0035] In some embodiments, the exothermic chamber 130 comprises a bottom inlet 132. In some embodiments, the bottom inlet 132 is configured to couple the combustion zone 131 of the exothermic chamber 130 to a source of combustion air. Thus, oxygen in the combustion air can burn fuel in the combustion zone 131 of the exothermic chamber 130. In some embodiments, the exothermic chamber 130 is configured to develop a fluidized bed in the combustion zone 131. Thus, the bed of the exothermic chamber 130 in the combustion zone 131 can be fluidized. During combustion, feedstock can float on the fluidized bed in the combustion zone 131 of the exothermic chamber 130. Thus, the feedstock can be evenly distributed around the at least one heat-transfer rod 140. In an embodiment of the gasification device that comprises a plurality of heat-transfer rods 140, the feedstock can be evenly distributed between the plurality of heat-transfer rods 140.

[0036] In some embodiments, the exothermic chamber 130 comprises a bottom outlet 134. In some embodiments, the bottom outlet 134 is configured to release ash from the exothermic chamber 130. At least one effect can be that the residue of feedstock after combustion, gasification and pyrolysis is removed from the gasification device 100.

[0037] In some embodiments, the exothermic chamber 130 is configured to combust a variety of fuels. The combustion zone 131 of the exothermic chamber 130 may be configured as a multi-fuel combustion zone capable of receiving and combusting the variety of fuels and/or configured with a multi-fuel burner, so that the variety of fuels can be selectively provided to the combustion zone 131 of the exothermic chamber 130. Feedstock can be anything that comprises carbon and burns. For example, feedstock can comprise one or more from a group of fermentable, biomass-containing residual materials consisting of sewage sludge, biowaste or food waste, farm manure (liquid manure, dung),

previously unused plants as well as plant parts (for example catch crops, plant residues and the like), specifically cultivated energy crops (renewable raw materials). Preferably, an amount of corrosive components, such as chloride or sulfide, in the feedstock is kept low or even to a minimum. One effect of keeping the amount of corrosive components in the feedstock low is that the corrosion of the chamber is reduced.

[0038] In particular, the exothermic chamber 130 can be configured to receive feedstock for combustion in the combustion zone 131 of the exothermic chamber 130. For example, the feedstock includes solid phase feedstock and/or gaseous/vaporized phase feedstock. Thus, the exothermic chamber 130 can be used to perform combustion on any phase of feedstock, thereby reducing, for example, a need for additional fuel for combustion. Further, a need for filters can be reduced that are conventionally used to separate feedstock suitable for combustion from other material. In some embodiments, the combustion zone 131 of the exothermic chamber 130 is adapted to receive process products and/or by-products of gasification obtained from pyrolysis performed in the second endothermic chamber 110. In some embodiments, the combustion zone 131 of the exothermic chamber 130 is adapted to receive products and/or by-products of gasification performed in the first endothermic chamber 110. Thus, use of feedstock in the gasification device 100 can be optimized.

[0039] In some embodiments, the combustion zone 131 of the exothermic chamber 130 can be configured to combust additional feedstock obtained directly from another source, like intermediate products of a pyrolysis reaction obtained from the external source 170 to produce process heat. Thus, a need to provide feedstock for combustion from the first endothermic chamber 110 and the second endothermic chamber 120 to the combustion zone 131 of the exothermic chamber 130 is reduced. Thus, the efficiency of the gasification device 100 to produce process heat is improved.

[0040] In some embodiments, the exothermic chamber 130 can be adapted to allow a variation of a configuration according to operational parameters of the gasification device 100, like the input of feedstock from any one or any combination of the inlet 172 and the inlet 174, temperature, pressure etc.

[0041] In some embodiments, the first endothermic chamber 110 of the gasification device 100 is adapted to perform an allothermic gasification process. In particular, the first endothermic chamber 110 can be configured as a reformer. For example, the first endothermic chamber 110 can comprise a multi-material allothermic gasification-reforming reactor. At least one effect can be that the gasification can be performed inside the gasification device 100 directly upon pyrolysis. In some embodiments, the first endothermic chamber 110 is configured to develop a fluidized bed. An effect of making the bed of the first endothermic chamber 110 fluidized can be, as the feedstock floats on the bed of the first endothermic chamber 110, that the feedstock is evenly distributed between the rods 140.

[0042] In some embodiments, the first endothermic chamber 110 comprises a reformer inlet 116. The reformer inlet 116 is configured to receive steam. The reformer inlet 116 couples the first endothermic chamber 110 to a source of steam. At least one effect of steam in the first endothermic chamber 110 can be that the steam is used to produce gas preferably, but not limited to, synthesis gas ($\text{CO}+\text{H}_2$), i.e.,

$\text{CH}_4+\text{H}_2\text{O}\rightleftharpoons\text{CO}+3\text{H}_2$. In some embodiments, the first endothermic chamber 110 comprises a first outlet 114. In some embodiments, the first endothermic chamber 110 further comprises a second outlet 115. An effect can be to transfer the feedstock from the first endothermic chamber 110 to the exothermic chamber 130.

[0043] In some embodiments, the first endothermic chamber 110 comprises a gas outlet 112. The gas outlet 112 is configured to supply the produced gas after, preferably synthesis gas combination of carbon monoxide and hydrogen, outside the gasification device 100. In some embodiments, the gas outlet 112 is coupled with a gas container (not shown) to supply the produced gas, preferably synthesis gas combination of carbon monoxide and hydrogen, outside the gasification device 100 directly to the container. At least one effect can be that the gas outlet 112 couples the first endothermic chamber 110 to the gas container (not shown) for further applications, like running engines.

[0044] In some embodiments, the first endothermic chamber 110 performs gasification process on feedstock to produce a gas. In some embodiments, the first endothermic chamber 110 is configured to perform gasification process on feedstock to produce a synthesis gas $\text{CO}+\text{H}_2$ or a producer gas $\text{CO}+\text{H}_2$ or CO_2+H_2 . In some embodiments, the first endothermic chamber 110 performs gasification process on feedstock to produce a hydrogen rich synthesis gas. The first endothermic chamber 110 operates at a temperature of greater than 700°C . or in the range of from 700°C . to 1100°C . Thus, gasification of the biodegradable material can take place inside the first endothermic chamber 110 by reacting the feedstock material at temperatures in an endothermic reaction.

[0045] In some embodiments, the first endothermic chamber 110 may be modified according to the operational parameters of the gasification device 100, like temperature, pressure etc.

[0046] The second endothermic chamber 120 is adapted to have the feedstock perform endothermic break-down reactions. In particular, the second endothermic chamber 120 is configured as a pyrolysis reactor. In some embodiments, the second endothermic chamber 120 comprises an inlet 122. The inlet 122 is configured to receive feedstock from a storage outside the gasification device 100. In some embodiments, the second endothermic chamber 120 comprises an upper outlet 124. In some embodiments, the second endothermic chamber 120 comprises a lower outlet 125. Thus, feedstock can be transferred from the second endothermic chamber 120 to the exothermic chamber 130. In some embodiments, the second endothermic chamber 120 is configured to develop a fluidized bed. Thus, solid particles of feedstock can flow in the fluidized bed, i.e., the solid and fluid parts of feedstock can be transported in the fluid within the gasification device 100.

[0047] In some embodiments, the second endothermic chamber 120 is configured to perform thermal oxygen-free decomposition on the feedstock to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The second endothermic chamber 120 is configured to operate at a temperature of equal to or above 350°C ., for example in the range of from 350°C . to 600°C . Thus, break-down of feedstock into various states of matter suitable for gasification in the first endothermic chamber 110 and for combustion in the exothermic chamber 130 can be performed in the second endothermic chamber 120.

[0048] In some embodiments, a first channel 181 is connected between the external source 170 of feedstock to the inlet 172 of the exothermic chamber 130. In some embodiments, the first channel 181 is further connected between the first outlet 114 of the first endothermic chamber 110 and the inlet 172 of the exothermic chamber 130. In some embodiments, the first channel 181 is also connected between the upper outlet 124 of the second endothermic chamber 120 and the inlet 172 of the exothermic chamber 130. Using the channel 181, feedstock can be transferred from the external source 170, the first endothermic chamber 110 and/or the second endothermic chamber 120 to the exothermic chamber 130 for combustion. In some embodiments, the first channel 181 is connected between the upper outlet 124 of the second endothermic chamber 120 and the first outlet 114 of the first endothermic chamber 110. Using the channel 181, feedstock can be transferred, for example, from the second endothermic chamber 120 to the first endothermic chamber 110.

[0049] In some embodiments, the gasification device 100 further comprises a second channel 182 that is connected between the inlet 174 to the exothermic chamber 130 and the second outlet 115 of the first endothermic chamber 110. The second channel 182 can further be connected between the inlet 174 to the exothermic chamber 130 and the lower outlet 125 of the second endothermic chamber 120. Using the second channel 182, feedstock can be transferred from the external source 170, the first endothermic chamber 110 and/or the second endothermic chamber 120 to the exothermic chamber 130. In some embodiments, the second channel 182 is connected between the lower outlet 125 of the second endothermic chamber 120 and the second outlet 115 of the first endothermic chamber 110. Using the second channel 182, feedstock can be transferred from the second endothermic chamber 120 to the first endothermic chamber 110 for gasification. Further using the second channel 182, feedstock can also be transferred from the second endothermic chamber 120 to the combustion zone 131 of the exothermic chamber 130.

[0050] In some embodiments, the inlet 172 is configured to receive feedstock in solid, liquid, decomposed and/or pyrolysis feedstock from the external source 170. In some embodiments, the inlet 172 is configured to receive feedstock in solid, liquid state, decomposed feedstock and/or residue of gasification from the first outlet 114 of the first endothermic chamber 110 for combustion. In some embodiments, the inlet 172 is configured to receive feedstock in gaseous and/or vaporized state from the upper outlet 124 of the second endothermic chamber 120 for combustion. At least one effect can be is to feed the exothermic chamber 130 with the feedstock for combustion. Yet another effect can be is to remove the residual feedstock after gasification from the gasification device 100.

[0051] In some embodiments, the first outlet 114 is configured to receive the feedstock in solid, liquid, decomposed and/or pyrolysis feedstock from the external source 170 for gasification. In some embodiments, the first outlet 114 is configured to receive pyrolysis feedstock in gaseous and/or vaporized state from the upper outlet 124 of the second endothermic chamber 120 for gasification. Thus, additional feedstock from the external source 170 can be fed into the first endothermic chamber 110 to produce a high yield of gas, preferably a high yield of synthesis gas (CO+H₂). Further, upon pyrolysis in the second endothermic chamber

120, processed or decomposed feedstock can be provided from the second endothermic chamber 120 to the first endothermic chamber 110, in particular, to perform gasification of the processed or decomposed feedstock.

[0052] In some embodiments, the inlet 174 is configured to receive feedstock in gaseous or vaporized state from the second outlet 115 of the first endothermic chamber 110 for combustion. In some embodiments, the inlet 174 is configured to receive feedstock in solid, liquid, decomposed or pyrolysis feedstock from the lower outlet 125 of the second endothermic chamber 120. In some embodiments, the second outlet 115 is configured to receive the feedstock in solid, liquid, decomposed and/or pyrolysis form from the lower outlet 125 of the second endothermic chamber 120 for gasification. The second endothermic chamber 120 being configured with the lower outlet 125 enables removal of pyrolysis ash from the second endothermic chamber 120. Further, the first endothermic chamber 110 can be fed from the second endothermic chamber 120 with feedstock in solid and/or fluid phase for gasification.

[0053] In some embodiments, the gasification device 100 comprises an exhaust outlet 152 which is provided, for example, in an upper wall portion of the gasification device 100. The exhaust outlet 152 is adapted to release waste heat from the gasification device 100.

[0054] In the gasification device according to the example illustrated in FIG. 1, the plurality of heat-transfer rods 140 can comprise one or more of a solid heat pipe, a hollow heat pipe and/or a convection heat pipe. In some embodiments, the plurality of heat-transfer rods are provided each as a solid heat pipe. In some embodiments, the plurality of heat-transfer rods are provided each as a hollow heat pipe. In some embodiments, the plurality of heat-transfer rods are provided each as a convection heat pipe.

[0055] Now, an exemplary heat-transfer rod comprised in the plurality of heat-transfer rods 140 will be described. The exemplary heat-transfer rod 140 comprises metal, for example, steel and/or a steel alloy. In some embodiments, the heat-transfer rod 140 comprises a core and a shell that, at least in a portion of the heat-transfer rod that is located in the combustion zone 131 of the exothermic chamber 130, encloses the core. The core is comprised of core material that differs from shell material. The shell is comprised of shell material that has a higher melting point than the core material. For example, the shell material can be steel, in particular, stainless-steel, while the core material can be copper, a copper alloy, aluminum, or an aluminum alloy. The exemplary heat-transfer rod 140 is configured to withstand a temperature of more than 1100° C. More particularly, the exemplary heat-transfer rod 140 is designed to withstand a temperature in the range of from 800° C. to 1000° C. In some embodiments, the heat-transfer rod 140 comprises material that has a melting point sufficiently above 1100° C. so that the heat-transfer rod 140 keeps its structure rather than bending or otherwise deforming or disintegrating when exposed to a temperature that is prevalent in the combustion zone 131 during operation of the exothermic chamber 130.

[0056] In some embodiments, the exemplary heat-transfer rod comprises a hollow heat pipe. A cavity inside the hollow heat pipe can be filled with medium suitable to perform convective heat transport. In some embodiments, the hollow heat pipe is filled with a working fluid. In some embodiments the cavity inside the hollow heat pipe is provided with an internal structure adapted to optimize convective heat

flow and delivery of heat at predetermined portions of the heat-transfer rod to the wall of the hollow heat pipe.

[0057] The exemplary heat-transfer rod is adapted to withstand a temperature of more than 1100° C. In some embodiments, the exemplary heat-transfer rod is adapted to withstand a temperature in the range from 800° C. to 1000° C., while transferring heat from the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110 and the second endothermic chamber 120.

[0058] In some embodiments, the exemplary heat-transfer rod 140 comprises a solid heat pipe. The solid heat pipe is made of a metallic or an alloy material. The solid heat pipe can have a smaller diameter than a hollow heat pipe. A hollow heat pipe, to achieve a same heat transfer, may require a larger diameter for filling the rod with fluid and for having the fluid flow inside the hollow heat pipe. Thus, the solid heat pipe reduces the need of extra resources to achieve the same or even better efficiency for heat transfer.

[0059] In some embodiments, as in the example illustrated in FIG. 1, the plurality of heat-transfer rods 140, arranged inside the exothermic chamber 130, extend through the first endothermic chamber 110 and into the second endothermic chamber 120. In some embodiments (not shown), the plurality of heat-transfer rods comprises a first set of at least one heat-transfer rod that extends into the first endothermic chamber 110. Further, the plurality of heat-transfer rods comprises a second set of at least one heat-transfer rod that extends into the second endothermic chamber 120. Thus, the plurality of heat-transfer rods 140 thermally couples the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110 and/or to the second endothermic chamber 120.

[0060] The plurality of heat-transfer rods 140 extend from the combustion zone of the exothermic chamber 130. The plurality of heat-transfer rods 140, in a portion provided in the combustion zone 131, are configured to withstand a temperature of at least 1100° C. The plurality of heat-transfer rods 140 are configured to transport heat from the combustion zone 131 to the first endothermic chamber 110, wherein the first endothermic chamber 110 is adapted for operation at a temperature in the range of from 700° C. to 1100° C. The heat-transfer rods 140 are configured to further transport heat to the second endothermic chamber 120, wherein the second endothermic chamber 120 is adapted to operate at a temperature in the range of from 350° C. to 600° C. Thus, the plurality of heat-transfer rods 140 can transfer process heat generated in the combustion zone 131 of the exothermic chamber 130 to the first endothermic chamber 110 and to the second endothermic chamber 120. In particular, heat can efficiently be transferred from the combustion zone 131 to the inside of the first endothermic chamber 110 and on from the inside of the first endothermic chamber 110 to the inside of the second endothermic chamber 120. In some embodiments, the heat-transfer rods 140 are provided with multiple segments that are stacked on one another to form the respective heat-transfer rod.

[0061] Accordingly, the gasification device 100 is configured to transfer the process heat not only by radiation and by convection from the exothermic chamber 130 to the first endothermic chamber 110 and/or to the second endothermic chamber 120, but also by conduction of heat in the heat-transfer rods 140. Convection provides for process heat, generated from the exothermic chamber 130 and rising by convection in the gasification device 100, to heat an outside

of the first endothermic chamber 110 and/or an outside of the second endothermic chamber 120. In contrast, conduction of heat in the plurality of heat-transfer rods 140 provides for the inside of the first endothermic chamber 110 and/or the inside of the second endothermic chamber 120 to be heated. Thus, the rate of heat transfer inside the gasification device 100 is increased, whereby the gasification device 100 can be more efficiently perform gasification than a conventional device.

[0062] In operation of an embodiment of the gasification device as illustrated in FIG. 1, combustible material is fed to the combustion zone 131 of the exothermic chamber 130. The combustible material can comprise feedstock in the form of pyrolysis vapors, gases and/or pyrolysis coke or ash provided from the first endothermic chamber 110 and/or from the second endothermic chamber 120. The combustible material is combusted in the combustion zone 131 of the exothermic chamber 130 and generates process heat, for example, at a temperature of above 1100° C.

[0063] From the combustion zone 131, process heat rises up to the first endothermic chamber 110 which, for example, operates at a temperature in a range of from 700° C. to 1100° C. In the first endothermic chamber 110, gasification of feedstock takes place to produce a hydrogen rich synthesis gas. In some embodiments, the gasification of feedstock produces a synthesis gas $\text{CH}_4 + \text{H}_2\text{O} = \text{CO} + 3 \text{H}_2$. From the first endothermic chamber 110, process heat rises up further to the second endothermic chamber 120 which, for example, operates at a temperature in a range of from 350° C. to 700° C. In the second endothermic chamber 120, feedstock is thermally decomposed without oxygen to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The feedstock is broken down into various states of matter suitable for gasification in the first endothermic chamber 110 and/or for combustion in the exothermic chamber 130.

[0064] In some embodiments, process heat rises up convectively from the exothermic chamber 130 to the outside of the first endothermic chamber 110 and to the outside of the second endothermic chamber 120. Thus, process heat is provided to the first endothermic chamber 110 and to the second endothermic chamber 120.

[0065] Further, the plurality of heat-transport rods 140 conduct process heat from the combustion zone 131 into the first endothermic chamber 110 and into the second endothermic chamber 120. More particularly, the plurality of heat-transfer rods 140, extending from the combustion zone 131 of the exothermic chamber 130 to the second endothermic chamber 120, and having a lower temperature at the second endothermic chamber 120 than in the combustion zone 131 of the exothermic chamber 130, conduct heat generated in the combustion zone 131, following a temperature gradient along the plurality of heat-transfer rods 140, to the second endothermic chamber 120. In the first endothermic chamber 110, the reformer process uses process heat from the combustion zone 131 to generate synthetic gas. In the second endothermic chamber 120, the pyrolysis uses process heat from the combustion zone 131 to break-down feedstock for use in the reformer process performed in the first endothermic chamber 110.

[0066] In some embodiments, the gasification device 100 further comprises a steam generator (not shown in FIG. 1) and the at least one heat-transfer rod 140 is further adapted to transfer the process heat from the exothermic chamber 130 to the steam generator (not shown in FIG. 1). The steam generator is configured to operate at a temperature of greater

than 120° C. or a temperature range from 120° C. to 350° C. The at least one heat-transfer rod **140** extending from the exothermic chamber **130** with a temperature of at least 1100° C. cools down to the temperature in the range of from 120° C. to 350° C., before contacting the steam generator. The process heat is dissipated to the first endothermic chamber **110** operating at the temperature in the range of from 700° C. to 1100° C., and to the second endothermic chamber **120** operating at the temperature in the range of from 350° C. to 700° C. At least one effect can be that the residual process heat of the at least one heat-transfer rod **140** and/or heat rising in the exothermic chamber **130** is further utilized for generating steam internally in the device.

[0067] FIG. 2 is a drawing that schematically illustrates a sectional view of a gasification device **200** for gasification of feedstock according to some embodiments that implements a steam generator. The illustrated embodiment is analogous to the embodiments defined above with reference to FIG. 1. Accordingly, the gasification device **200** comprises an exothermic chamber **230** that is provided with a combustion zone **231**. Further, the gasification device **200** comprises a first endothermic chamber **210**, a second endothermic chamber **220** and a plurality of heat-transfer rods **240**. In addition, the gasification device **200** comprises a steam generator **260**. Using the plurality of heat-transfer rods **240**, process heat generated in the combustion zone **231** of the exothermic chamber **230** can thus be conducted to the first endothermic chamber **210**, to the second endothermic chamber **220** and to the steam generator **260**. At least one effect of the steam generator **260**, when operated to generate steam, can be a reduction of steam that needs to be provided from a source external to the gasification device **200**.

[0068] In some embodiments, at least one of the first endothermic chamber **210**, the second endothermic chamber **220** and the steam generator **260** is arranged inside the exothermic chamber **230** of the gasification device **200**. For example, as illustrated in FIG. 2, the second endothermic chamber **220** and the first endothermic chamber **210** are arranged inside the exothermic chamber **230**, above the combustion zone **231**, in a stack above one another, while the steam generator **260** is provided above the stack. In the exemplary embodiment, a ceiling wall **239** of the exothermic chamber **230** forms a floor wall of the steam generator **260**, whereby the steam generator **260** is thermally integrated with the gasification device **200**. Thus, the second endothermic chamber **220**, the first endothermic chamber **210** and the steam generator **260** can use process heat generated in the exothermic chamber **230**. Thus, the first endothermic chamber **210**, the second endothermic chamber **220** and the steam generator **260** can carry out the endothermic reactions and steam generation, respectively, using process heat produced in the exothermic chamber **230**.

[0069] The steam generator **260** is coupled, via a steam outlet **269** connected by a steam channel **283** to a reformer inlet **216**, to the first endothermic chamber **210**. At least one effect can be that the steam which is produced within the gasification device **200** can be used in the first endothermic chamber **210** to produce a synthesis gas ($\text{CO}+\text{H}_2$), i.e., $\text{CH}_4+\text{H}_2\text{O}\rightleftharpoons\text{CO}+3\text{H}_2$.

[0070] In an exemplary embodiment, the steam generator **260** comprises a water preheater **262**, a steam producer **263** and a steam super-heater **264**. The water preheater **262** is configured to use process heat to heat the water to a temperature in the range of from 120° C. to 500° C. In some

embodiments, the water preheater **262** is configured to use process heat to heat the water to a temperature in the range of from 180° C. to 400° C. In some embodiments, the water preheater **262** is configured to use process heat to heat the water to a temperature in the range of from 240° C. to 300° C. The heated water can then be transferred to the steam producer **263**. In some embodiments, the water is kept as water above atmospheric pressure. In some embodiments, the water is kept at atmospheric pressure, whereby the process heat is used to convert the water to steam.

[0071] In some embodiments, the water preheater **262** of the steam generator **260** is coupled, via a water inlet **265**, to an external water source **279** for intake of water. In some embodiments, process heat can be used by the water preheater **262** to preheat the water taken in for use in the generation of steam. In particular, in an embodiment, the steam generator **260** is configured to release process heat from the gasification device **200**, via a first exhaust outlet **267**, to the external water source **279**. Thus, heat of exhaust gases can preheat water of the external water source **279**.

[0072] In some embodiments, the steam super-heater **264** is configured to produce steam using heat of synthesis gas produced by the first endothermic chamber **110**.

[0073] In some embodiments, a gas outlet **212** is configured to supply synthesis gas to the outside of the gasification device **200**. In some embodiments, the gas outlet **212** is coupled to the steam generator **260** to supply the synthesis gas from the first endothermic chamber **210** to the steam generator **260** for producing steam. In some embodiments, the gas outlet **212** is coupled to the super-heater **264** and configured to supply gas from the first endothermic chamber **210** to the super-heater **264** for producing steam. In some embodiments, the gas outlet **212** is coupled to the steam generator **260** to supply some part of the gas to the steam generator **260** for producing steam and some part to a gas container (not shown) for other applications of the synthesis gas, such as for running an engine. In some embodiments, the gas outlet **212** is coupled to a gas container (not shown) for supply of the synthesis gas outside the gasification device **200** to the container (not shown). At least one effect can be that the gas outlet **212** couples the first endothermic chamber **210** to the steam generator **260** and/or to the gas container (not shown) so that the gas usage is maximized and no gas is leaked or wasted to the surroundings.

[0074] The heat-transport rods **240** extend from the bottom of the exothermic chamber **230** and are adapted to dissipate process heat to the inside of the first endothermic chamber **210**, to the inside of the second endothermic chamber **220** and to the inside of the steam generator **260**. At least one effect can be that the plurality of heat-transfer rods **240** can conduct heat from the combustion zone **231** of the exothermic chamber **230** to the inside of the first endothermic chamber **210**, to the inside of the second endothermic chamber **220** and/or to the inside of the steam generator **260**.

[0075] In some embodiments, the plurality of heat-transfer rods **240** extends into the steam producer **263** for generating steam while using process heat dissipated from the plurality of heat-transfer rods **240**. The steam producer **263** is configured to convert the heated water to steam using process heat conducted by the plurality of heat-transfer rods **240**.

[0076] The inlet **222** is configured to receive feedstock. In some embodiments, the feedstock is supplied from any storage outside of the gasification device **200**. The second

endothermic chamber 220 performs pyrolysis on the feedstock. In particular, the second endothermic chamber 220 performs thermal oxygen-free decomposition on the feedstock to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The second endothermic chamber 220 operates at a temperature of greater than 350° C. or in the range of from 350° C. to 600° C. At least one effect of the second endothermic chamber 220 can be to.

[0077] In some embodiments, the second endothermic chamber 220 may be modified according to the operational parameters of the gasification device 200, like temperature, pressure etc.

[0078] The steam generator 260 comprises a water preheater 262, a steam producer 263 and/or a steam superheater 264. At least one effect can be that the exhaust gas from the gasification device 200 is used to run the steam generator 260, when the steam generator 260 is integrated with the gasification device 200.

[0079] Accordingly, the gasification device 200 is configured to transfer the process heat from the exothermic chamber 230 to the first endothermic chamber 210, to the second endothermic chamber 220 and/or to the steam generator 260 by two channels. The first channel is to heat the outside of the first endothermic chamber 210, the second endothermic chamber 220 and/or the steam generator 260 by the process heat, generated from the exothermic chamber 230, rising in the gasification device 200. The second channel is to heat the inside of the first endothermic chamber 210, the second endothermic chamber 220 and/or the steam generator 260 by the heat-transfer rods extending from the exothermic chamber 230. At least one effect of both the channels of heat transfer can be to increase the rate of heat transfer inside the device, whereby making the device more efficient for the process of gasification.

[0080] In operation of an embodiment of the gasification device as illustrated in FIG. 2, combustible material is fed to the combustion zone 231 of the exothermic chamber 231. The combustible material can comprise feedstock in the form of pyrolysis vapors, gases and/or pyrolysis coke or ash provided from the first endothermic chamber 210 and/or from the second endothermic chamber 220. The combustible material is combusted in the combustion zone 231 of the exothermic chamber 230 and generates process heat, for example, at a temperature of above 1100° C.

[0081] From the combustion zone 231, process heat rises up to the first endothermic chamber 210 which, for example, operates at a temperature in a range of from 700° C. to 1100° C. In the first endothermic chamber 210, gasification of feedstock takes place to produce a hydrogen rich synthesis gas. In some embodiments, the gasification of feedstock produces a synthesis gas $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3 \text{H}_2$. One example of the produced synthesis gas is a gas comprising H_2 and CO components. In another example, the produced synthesis gas comprises CO_2 , O_2 , CH_4 and/or N_2 components. From the first endothermic chamber 210, process heat rises up further to the second endothermic chamber 220 which, for example, operates at a temperature in a range of from 350° C. to 700° C. In the second endothermic chamber 220, feedstock is thermally decomposed without oxygen to produce pyrolysis gases and vapors, and pyrolysis coke/ash. The feedstock is broken down into various states of matter suitable for gasification in the first endothermic chamber 210 and for combustion in the exothermic chamber 230.

[0082] From the second endothermic chamber 220, process heat rises up still further to the steam generator 260 which, for example, operates at a temperature in a range of from 120° C. to 350° C.

[0083] In some embodiments, the process heat rises up convectively from the exothermic chamber 230 to the outside of the first endothermic chamber 210, to the outside of the second endothermic chamber 220 and to the steam generator 260. Thus, the process heat warms up the first endothermic chamber 210, the second endothermic chamber 220 and the steam generator 260.

[0084] Further, the plurality of heat-transport rods 240 conduct process heat from the combustion zone 231 into the first endothermic chamber 210, into the second endothermic chamber 220 and into the steam generator 260. More particularly, the plurality of heat-transfer rods 240, extending from the combustion zone 231 of the exothermic chamber 230 to the steam generator 260, and being colder at the steam generator 260 than in the combustion zone 231, conduct heat generated in the combustion zone 231 of the exothermic chamber 230, following a temperature gradient along the plurality of heat-transfer rods 240, to the steam generator 260. The steam generator uses the process heat from the exothermic chamber 230 in the generation of steam. The steam is directed from the steam generator 260 into the first endothermic chamber 210 where the steam is used to perform the reformer process to produce synthesis gas.

[0085] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. For example, while in the examples illustrated in FIG. 1 and in FIG. 2 the gasification device comprises a plurality of heat-transfer rods, in another example (not shown) the gasification device merely comprises a single heat-transfer rod.

[0086] Although some drawings may be provided with exemplary dimensional statements, it should be understood that such statements are merely exemplary. Neither can any such statements be understood to be consistent from one drawing to another, nor should the statements be understood to limit the scope of the present disclosure to the stated dimensions or any combination or ratio thereof. The disclosure of dimensions should be understood merely to state an order of magnitude according to some embodiments. In particular, the invention can be implemented using, other orders of magnitude, other dimensions, other ratios of dimensions. Further, it should be understood that drawings are not drawn to scale.

Molecules

[0087] The implementations herein are described in terms of exemplary embodiments. However, it should be appreciated that individual aspects of the implementations may be separately claimed and one or more of the features of the various embodiments may be combined.

1. A gasification device, the gasification device comprising:
 - an exothermic chamber provided with a combustion zone that is configured to perform combustion to produce process heat,

- a first endothermic chamber adapted to perform gasification of feedstock; and
- a second endothermic chamber adapted to subject feedstock to an endothermic break-down reaction;
- wherein the gasification device is configured such that the first endothermic chamber uses combustion process heat to perform the gasification process, and
- wherein the gasification device is configured such that the second endothermic chamber uses combustion process heat to subject feedstock to the endothermic break-down reaction;
- the gasification device further comprising at least one heat-transfer rod adapted to transfer combustion process heat from the combustion zone of the exothermic chamber to the first endothermic chamber and to the second endothermic chamber.
2. The gasification device of claim 1, wherein the at least one heat-transfer rod is arranged inside the exothermic chamber and extends into the first endothermic chamber and into the second endothermic chamber.
3. The gasification device of claim 1, wherein the at least one heat-transfer rod comprises a solid heat pipe, a hollow heat pipe or a convection heat pipe.
4. The gasification device for gasification of feedstock of claim 1, wherein at least two of the second endothermic chamber, the first endothermic chamber and the combustion zone of the exothermic chamber are arranged above one another.
5. The gasification device of claim 1, wherein at least one of the second endothermic chamber and the first endothermic chamber is arranged inside the exothermic chamber.
6. The gasification device of claim 1, wherein the first endothermic chamber is a reformer.

7. The gasification device of claim 6, wherein the reformer is configured to comprise a fluidized reformer-bedchamber.

8. The gasification device of claim 1, wherein the first endothermic chamber includes a reformer inlet configured to be coupled to a source of steam.

9. The gasification device of claim 8, the device further comprising a steam generator configured to receive water and to generate steam from the water, wherein the steam generator comprises a steam outlet connected to the reformer inlet.

10. The gasification device of claim 9, wherein the at least one heat-transfer rod is further adapted to transfer combustion process heat from the combustion zone of the exothermic chamber to the steam generator.

11. The gasification device of claim 1, wherein the second endothermic chamber is a pyrolysis reactor.

12. The gasification device of claim 11, wherein the pyrolysis reactor comprises a fluidized reactor-bedchamber.

13. The gasification device of claim 1, wherein the combustion zone of the exothermic chamber is configured to combust fuel to produce combustion process heat.

14. The gasification device of claim 1, wherein the exothermic chamber comprises an inlet configured to receive feedstock, and wherein the combustion zone of the exothermic chamber is configured to combust the feedstock.

15. The gasification device of claim 1, wherein the exothermic chamber further comprises a bottom inlet configured to be coupled to an oxygen source, and/or wherein the exothermic chamber further comprises a bottom outlet configured to release ash from the exothermic chamber.

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